Charotar University of Science and Technology

M.Sc. (Advanced Organic Chemistry) Semester I Examination, April 2011.

OC 702: Organic Chemistry-I

(Stereochemistry, reaction intermediates and mechanisms)

Date: 26th April, 2011 Time: 10.00 am to 10.30 am Max. Marks: 20

Part-A

(1) Formation of transition state is the characteristic of

(a) E1 reactions (b) E2 reactions

(c) S_N1 reactions (d) Both a and c

(2) Rate of unimolecular elimination reaction is dependent on ________.

(a) Concentration of nucleophile

(b) Concentration of substrate

(c) Concentration of both substrate and nucleophile

(d) None of these

(3) Acid catalyzed dehydration of alcohol follows

(a) E1 mechanism (b) E2 mechanism

(c) E1CB mechanism (d) S_N1 mechanism

(4) Aldehydes and ketones react with hydroxylamine to give

(a) Hydrazones (b) Phenylhydrazones

(c) Oximes (d) Semicarbazones

(5) Substituents such as _OH, -OR, _NHCOR or $\_NR_2$ activates the aromatic ring through_____.

(a) +I (inductive) effect (b) –I (inductive) effect

(c) +M (mesomeric) effect (d) None of these

(6) The correct order for stability of free radical is

(a) $Ph_2CH^{\cdot} > PhCH_2^{\cdot} > Ph_3C^{\cdot}$ (b) $PhCH_2^{\cdot} > Ph_2CH^{\cdot} > Ph_3C^{\cdot}$

(c) $Ph_3C^{\cdot} > Ph_2CH^{\cdot} > PhCH_2^{\cdot}$ (d) None of these

(7) In singlet and triplet state of carbene the carbon atom is _____ and ____ hybridized respectively.

(a) sp^2 and sp (b) sp and sp^2

(c) sp^2 and sp^3 (d) sp^3 and sp^2

(8) The basic requirement for carbocation to be stable is that it should be_________.

(a) Planar (b) Co-planar (c) Non planar (d) Tetrahedral

(9) Diels alder reaction involves__________

(a) Nucleophilic addition (b) Concerted mechanism

(c) Free radical mechanism (d) Electrophilic addition

(10) Nucleophilic addition is observed in _________.

(a) Ketones (b) Esters (c) Nitriles (d) All of these

(11) Nucleophilicity of aromatic ring can be increased by introduction of

(a) Nitro group (b) Methyl group

(c) Hydroxy group (d) Both b & c

(12) If position of two substituents are interchanged at an atom to give rise to a new stereoisomers, the atom at which the position of substituents are changed is called

(a) Chirogenic centre (b) Stereogenic centre

(c) Prochiral centre (d) Prostereo centre

(13) Number of stereoisomers in symmetrically substituted molecule having even number of carbon atoms can be calculated by the formula _______.

(a) 2^{n-1} (b) $2^{n-1} + 2^{n-2/2}$ (c) $2^{n-2/2}$ (d) $2^{n-1/2}$

(14) Give reaction for generation of carbene from diazirines.

(15) Give reaction for generation of acyl nitrene.

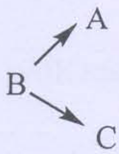
(16) Name any three reactions which involve formation of carbanion.

(17) What is role of Lewis acid catalyst in halogenations of aromatic ring?

(18) Define energy of activation.

(19) Consider the given reaction.

Free energy of product A is less than B and free energy of product C is greater than B. The free energy of activation for path B→A is higher than free energy of activation for B→C. Under such circumstances following is true.



(a) C is Kinetic control product and A is thermodynamic control product

(b) Both A and C are kinetic control product

(c) C is thermodynamic control product and A is Kinetic control product

(d) Both A and C are thermodynamic control product

(20) The reaction with no free energy of activation requirement at all, are known as

(a) Thermodynamically controlled reactions

(b) Kinetically controlled reactions

(c) Diffusion controlled reactions

(d) None of these

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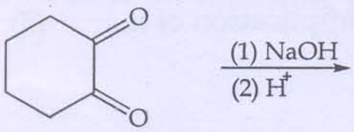
Part-B

Ques 1:

(I) Describe Baeyer Villiger oxidation with mechanism giving any two applications. (4)

(II) Write the end products for the following reactions. (Any six) (6)

(a)



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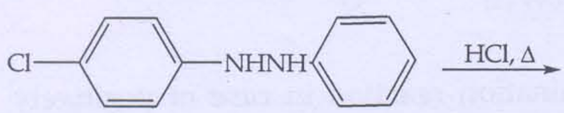

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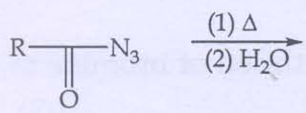

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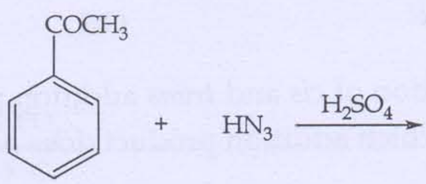

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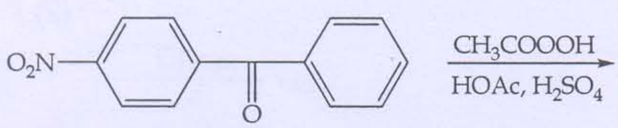

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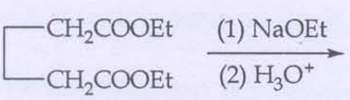

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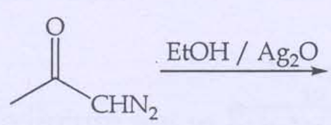

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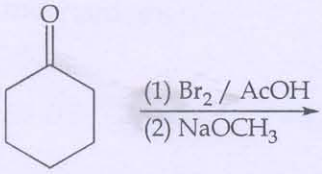
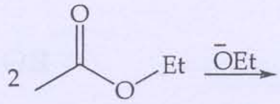


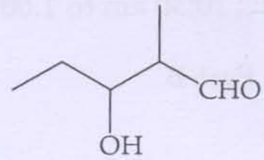
(i)



Ques 2:

(I) Explain stability of carbocations in terms of hyperconjugation and resonance giving suitable illustrations. (4)

(II) Explain the mechanism of Aldol condensation for the formation of given product. (3)



OR

(II) Describe mechanism of Sommelet-Hauser rearrangement and give any one application. (3)

(III) Write mechanism of Perkin reaction and also give one application of it. (3)

OR

(III) Describe mechanism of Curtius rearrangement with one application. (3)

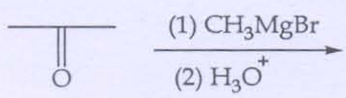
Ques 3:

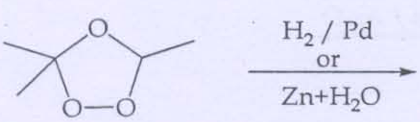
(I) Explain orientation of bimolecular elimination reaction in case of positively charged species showing mechanism. (3)

OR

(I) Explain formation of *cis* and *trans* addition products by addition of bromine to *cis*-2-butene. Which addition product does not form? (3)

(II) Complete the following reactions. (2)

(a) 

(b) 

(c) 

(d) $H_2C{=}CH_2 + KMnO_4 \xrightarrow[H_2O]{}$

(III) Explain formation of antimarkownikoff's product by addition of HBr to propene. (3)

OR

(III) Discuss addition of unsymmetrical reagent to unsymmetrical alkenes based on Markownikoff's rule. (3)

(IV) Discuss reactivity of addition reaction of alkenes Vs alkynes. (2)

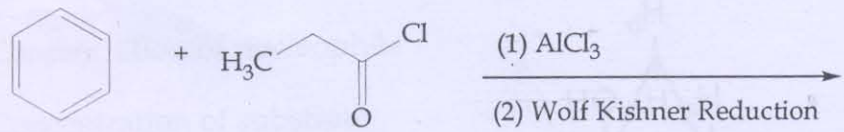
Ques 4:

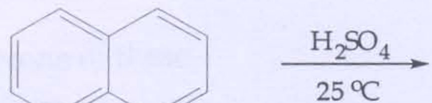
(I) What is unimolecular elimination reaction? Explain with mechanism. (3)

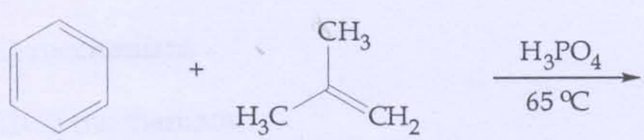
OR

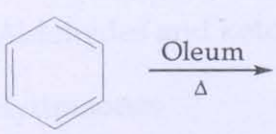
(I) Discuss stereochemistry of E2 reaction with suitable example. (3)

(II) Complete the following reactions. (2)

(a) 

(b) 

(c) 

(d) 

(III) State requirement of E1cB reaction. (2)

(IV) Give mechanism for formation of acylinium ion in Friedal Craft acylation and state disadvantages of Friedal Craft alkylation with illustration showing mechanism. (3)

OR

(IV) Discuss nitration of benzene with mechanism. Give reactions for nitration of phenol and toluene. (~~4~~) (3)

Ques 5:

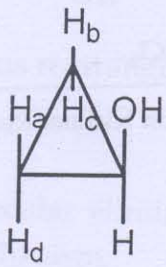
(I) Define homomorphic ligands and also explain stereoheterotopic and constitutional heterotopic ligands with example. (4)

(II) Why biphenyl compounds are chiral? Give the conditions to exhibit chirality for biphenyl compounds with example showing chiral and achiral compounds. (3)

OR

(II) What is chiral plane? State the conditions for allenes to be chiral and give one example of each chiral and achiral allenes. (3)

(III) In given compound, H_a and H_d are ________, H_b and H_c are ________, H_c and H_d are ________, H_a and H_b are ________. (2)



(IV) Define prosteregenic plane. (1)